Programming and Spreadsheet Modeling

Spreadsheet Engineering

Some graphics and datasets used in this presentation and accompanying MS Excel models are created by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering Overview

- The craft of modeling - art vs. science
- Audience and purpose
- Structure and functionality
- Understandable, modifiable and auditable
- Data, Parameters, Formulas
- Output and display

© 2026 W. Brian Lambert, PhD, All Rights Reserved

The Craft of Modeling
Follow guidelines, except when you shouldn't …

- Never hard code values in formulas
  - EVER
- *"Always use named ranges"*
  - Except when part of a formula uses relative references, or when …
- *"Always use tables instead of ranges"*
  - Except sometimes when importing sheets, or when …
- "*Spread functionality across sheets*"
  - Except when it corrupts the overall purpose of the model, or makes the user flip back and forth between sheets, or …

© 2026 W. Brian Lambert, PhD, All Rights Reserved

The Craft of Spreadsheet Modeling
What the model must do drives how to build it

- *"You should always use tables instead of ranges"*
  - That can causes issues with formulas when copying a sheet from one book to another
- *"Data should always be kept in a column format"*
  - Depends on where and what it's used for
  - If the table of formulas referencing the data is in row format you want the data shape to meet the formula shape
  - If it's on a user / summary sheet may need a row format to make the user interface right

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering

Requirements, Functionality and Structure

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering

- One part of the art in building models is to choose what's in and what's out.

"Everything should be made as simple as possible, but not simpler."
Albert Einstein

- Plato - never mistake the model for reality!

"All models are wrong, but some are useful"
George E. P. Box.

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Requirements: Who is the audience

- Yourself
- Management
- Peers
- Team including non-peers
- Users outside of immediate team-mates

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Requirements: What is the Purpose

- Custom calculator for current and future use
- One-off analysis
- Finite project
- Recurring analysis
- Implementable tool

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Requirements: Audience and Purpose

Purpose	Audience				
	Myself	Peers	Team	Management	Non-tech Users
Whiteboard	X				
One-off analysis					
Finite project					
Recurring analysis					
Custom Calculator					X
Implemented tool					X

Biggest risk, biggest reward
This can make you a name
And can be your nightmare ...
Did you sign up to be the help desk?

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering

Structure



© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
A model is not just a bunch of numbers and formulas

- A model includes data, text, adjustable parameters, calculations, visualizations, and documentation
- The intended audience and usage of the model are key factors in building a good model
- These two models are functionally identical
- Which would be better in a meeting and why?

Employee	Bob		1	Annual Bonus
Salary	$	30.0	2	5%
Bonus	$	1.5	3	Display Factor ($)
Total Pay	$	31.5	4	1
Employee	Salary		Bonus	Total Pay
Bob	$	30.0	$ 1.5	$ 31.5
Fred	$	55.0	$ 2.8	$ 57.8
Joe	$	25.0	$ 1.3	$ 26.3
TJ	$	60.0	$ 3.0	$ 63.0

Employee Compensation Model

Employee	Bob		1		Annual Bonus
Salary	$	30.0	2		5%
Bonus	$	1.5	3		Display Factor ($)
Total Pay	$	31.5	4		1

Employee	Salary	Bonus	Total Pay
Bob	$ 30.0	$ 1.5	$ 31.5
Fred	$ 55.0	$ 2.8	$ 57.8
Joe	$ 25.0	$ 1.3	$ 26.3
TJ	$ 60.0	$ 3.0	$ 63.0

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality

- A good model balances usability, logical structure, separation of components, auditability and **maintainability**
  - Useability drives everything
  - Auditability, maintainability, documentation
  - Data and input parameters
  - Formulas
  - Outputs and display
- Data, formulas, and summary results should be clearly identifiable
  - This can be easily done with a color scheme / cell style

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality: Auditability and maintainability

- Maintainability and auditability means that anyone can easily see the structure and function of your model

Maintainability:

- Assume the model will eventually be enhanced or modified
  - Unmaintainable models can require rework/rebuild for modest modifications
- You will likely be required to modify or add functionality long after you remember why you built it the way you did

Auditability:

- A manager may want to understand your model calculations
- **Sarbanes-Oxley & Internal Controls over Financial Reporting**
- If your Excel model produces numbers that modify a financial statement of a publicly traded company it is open to be audited by external auditors

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality: Data

- Consider both data sets and parameters as data
- Maintainability is **always** a concern
- Raw data sets should be protected from modification
  - If the model is limited to a spreadsheet, sometimes it's wise to keep a separate, dedicated "raw_data" or "orig_data" worksheet
- Data ranges usually should be named
- In large models, data should be kept on dedicated worksheets
- In smaller models this is not always practical
- For usability and context, data is often adjacent to or even intermixed with formulas
- Regardless of configuration, data must be clearly identified

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality: Inputs parameters

- NEVER hard code values into formulas – use parameters
- Parameters are known or assumed values used in the model
- If parameters can take on different values, use a lookup or data validation
- Watch out for pairs of parameters
  - Multiple assumptions may go together for a scenario
- Give meaningful names to ranges that contain parameters
- Should be color coded the same as data

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality: Formulas

- NEVER hard code values into formulas
  - Makes even models completely unauditable and unmaintainable
- Don't let it take a stupid-pill!
- Verify they work as you build
- Get familiar with formula auditing tools
- All of these formulas do the same thing – which is more maintainable?

  = B7/(B8-B9)

  = B7 / (B8 - B9)

  = fixed_cost / (sell_price – var_cost)
- Named ranges create a self-documenting model!
- Consider separate colors for cells that calculate values and just reference

© 2026 W. Brian Lambert, PhD, All Rights Reserved

Spreadsheet Engineering
Structure & Functionality: Output and displays

- What is the purpose of the model?
- Output and display of information is key if it's to be used for analysis
- Sloppy output ruins it all
  - Unformatted numbers and decimal places
  - Unaligned numbers or text, etc.
  - Formatting can take **forever**
  - Time is the enemy
- Appropriate type of chart to show the point and can be understood
  - "*I know it's an eye chart but it's showing…*"
- Again, color-coded cells / styles help with this tremendously

© 2026 W. Brian Lambert, PhD, All Rights Reserved